

Foundations of 4D/6D Object Capture for Virtual Environments



Human Models



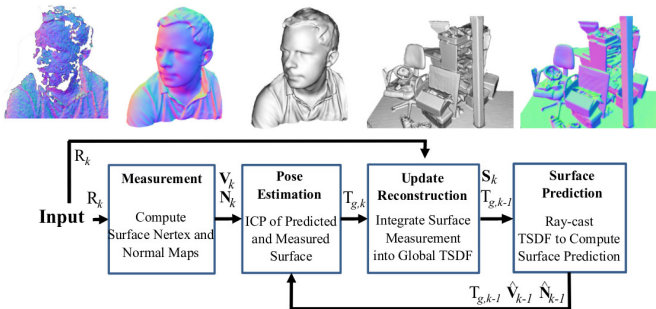
March 2, 2026

→ Reconstruction Methods



Today: Human Models

Question: Can we use reconstruction methods, like KinectFusion, to model humans?



Yes, if we can teach to handle deformations properly¹

→ "Only" need to adapt Pose Estimation from **single static pose** to **per-point trajectory**

¹R. A. Newcombe et al. "DynamicFusion: Reconstruction and Tracking of Non-rigid Scenes in Real-Time". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2015.

Given: Sequence of on-the-fly processed RGB-D images



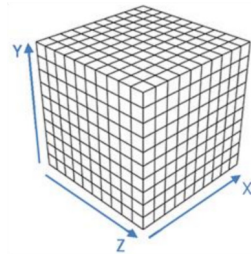
Idea: Use non-rigid version of Iterative Closest Point (ICP)

- Model deformations with both rotation and translation

$$T_{\text{warp}}(x) = \left[\begin{array}{c|c} R(x) & t(x) \\ \hline \mathbf{0} & 1 \end{array} \right]$$

→ Easier to handle complex motions although $2 \times$ parameters

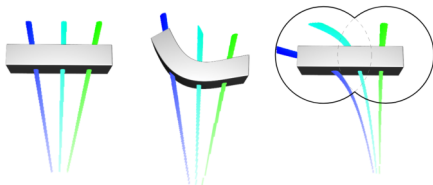
- Store one transformation $T_{\text{warp}}((i, j, k)^T)$ per grid cell $(i, j, k)^T \in \mathbb{Z}^3$
→ In practice, "sparse" set of nodes instead to reduce parameter count



We need a common space to perform the reconstruction → **Canonical Model** (usually 1st frame at $t = 0$)

Task:

1. Warp each cell from canonical model to current live frame



Live frame $t = 0$ Live Frame $t = 1$ Canonical $\mapsto$ Live

2. Use non-rigid ICP to update warp field
3. Fuse *projective TSDF* like in KinectFusion
4. Extract view from model via Sphere Tracing



Canonical Model



Canonical model warped into its live frame

Results:



Although the reconstruction can be performed in real time, it has shortcomings:

- ▶ **Not generalizable to other people** → Would need to capture each person with RGB-D camera
- ▶ **Not animatable** → Person would need to perform specific movement

→ Need a parameterized shape model of humans

Given: Dataset $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n \mid \mathbf{x}_i \in \mathbb{R}^d\}$

Goal: Linearly express samples $\mathbf{x}_i \in \mathbb{R}^d$ in $\mathcal{X}$ as closely as possible with lower number of parameters $k \ll d$

Find principal components, i.e. directions, $\{\mathbf{v}_j \in \mathbb{R}^d\}$ along which the variance in $\mathcal{X}$ is maximal

0. Start with zero-mean data $\hat{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$

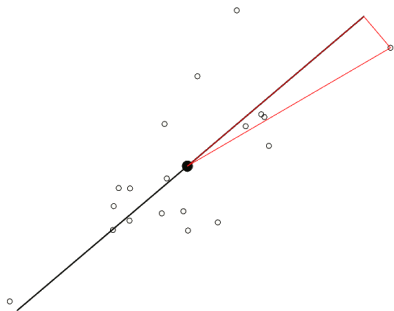
1. Compute $\mathbf{v}_1$ as eigenvector to largest eigenvalue of $\mathbf{C} \in \mathbb{R}^{d \times d}$:

$$\mathbf{v}_1 = \arg \max_{\mathbf{v} \in \mathbb{R}^d, \|\mathbf{v}\|=1} \sum_{i=1}^n \langle \hat{\mathbf{x}}_i \mid \mathbf{v} \rangle^2 = \arg \max_{\mathbf{v} \in \mathbb{R}^d, \|\mathbf{v}\|=1} \mathbf{v}^T \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \hat{\mathbf{x}}_i \hat{\mathbf{x}}_i^T \right)}_{=: \mathbf{C}} \mathbf{v}$$

2. Subtract any contribution in direction $\mathbf{v}_1$:

$$\hat{\mathbf{x}}_i \leftarrow \hat{\mathbf{x}}_i - \langle \hat{\mathbf{x}}_i \mid \mathbf{v}_1 \rangle \mathbf{v}_1$$

3. Repeat steps 1 + 2 to get $\mathbf{v}_2, \mathbf{v}_3, \dots$



Idea: In practice, this can be computed in a single pass

- Construct covariance matrix:

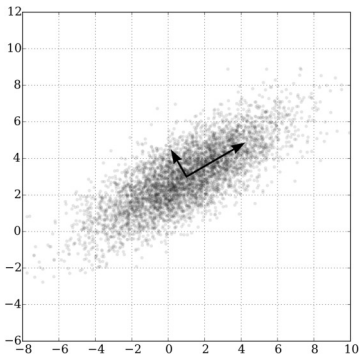
$$C = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$$

- Compute eigenvectors $(v_1, v_2, \dots, v_d)$ of $C \in \mathbb{R}^{d \times d}$
→ Sorted according to eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$

Now, we can exactly represent ($k = d$) or **approximate** ($k \ll d$) $\mathcal{X}$:

$$x(\alpha) = \bar{x} + \sum_{j=1}^d v_j \cdot \alpha_j$$

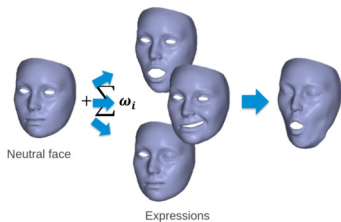
$$\approx \bar{x} + \sum_{j=1}^k v_j \cdot \alpha_j = \bar{x} + \left[\begin{array}{c|c|c} v_1 & \dots & v_k \end{array} \right] \cdot \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{bmatrix} = \bar{x} + V \cdot \alpha$$



Components v_j scaled by $\sqrt{\lambda_j}$

→ We can also "generate" new samples that follow distribution of $\mathcal{X}$

Application: Human faces²

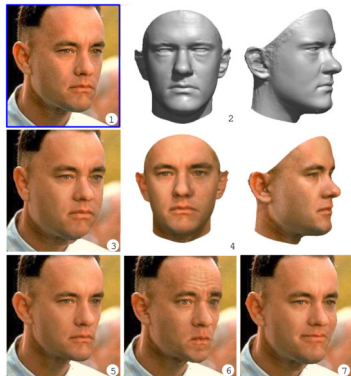


- Shape: Apply PCA on the geometry

$$\mathbf{S}_i = (X_1, Y_1, Z_1, X_2, Y_2, Z_2, \dots)^T \in \mathbb{R}^{3N}$$

- Color: Apply PCA on the texture

$$\mathbf{C}_i = (R_1, G_1, B_1, R_2, G_2, B_2, \dots)^T \in \mathbb{R}^{3N}$$



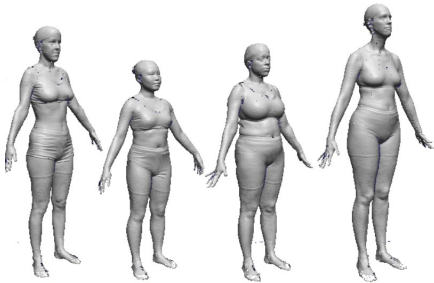
²V. Blanz and T. Vetter. "A Morphable Model for the Synthesis of 3D Faces". In: *Annual Conference on Computer Graphics and Interactive Techniques*. 1999.

In order to compute a human shape model, we first need some data to build upon

→ **C**ivilian **A**merican and **E**uropean **S**urface **A**nthropometry **R**esource (**CAESAR**) Project



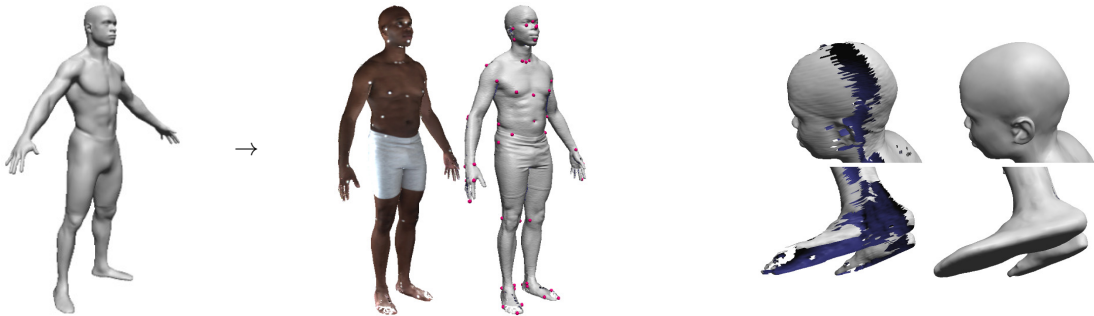
Cyberware 3D whole body scanner



- ▶ 3D scans of 125 male + female
- ▶ 74 white markers at anthropometric landmarks
- ▶ Demographic data (age, weight, ethnic group)

Goal: For humans, compute a PCA model of CAESAR dataset³

Start with artist-generated template model and align it to the scan



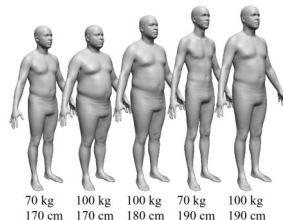
→ Vertices across each scan are now in correspondence for PCA

³B. Allen et al. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM Transactions on Graphics (TOG)* 22.3 (2003).

Problem: The principal components $\mathbf{v}_j \in \mathbb{R}^{3N}$ have no intuitive meaning

Idea: Define meaningful features $\mathbf{f} = (f_1, \dots, f_L)^T \in \mathbb{R}^L$:

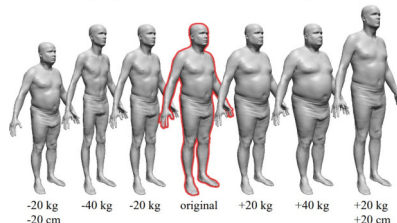
- ▶ Height
- ▶ Weight
- ▶ Age
- ▶ ...



Compute linear transformation:

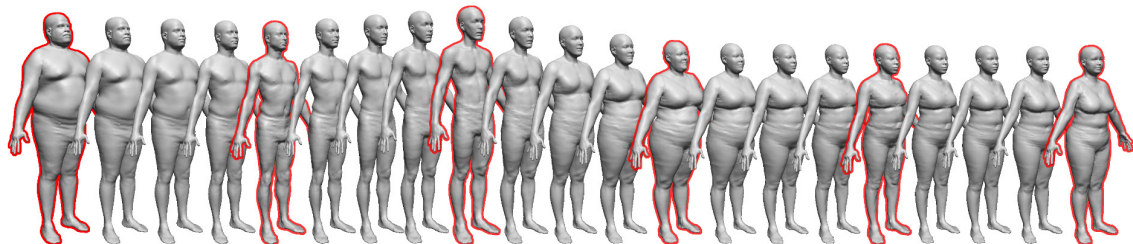
$$\mathbf{M} \cdot (\mathbf{f}, 1)^T = \boldsymbol{\alpha}$$

- ▶ Maps features $\mathbf{f}$ to PCA weights $\boldsymbol{\alpha}$
- ▶ Can compute delta vectors due to homogeneous coordinates



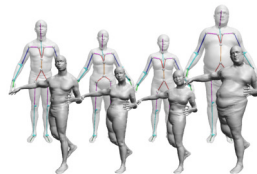
Results: Shape model

$$T_{\text{new}}(\mathbf{f}) = \overline{T} + V \cdot M \cdot (\mathbf{f}, 1)^T$$



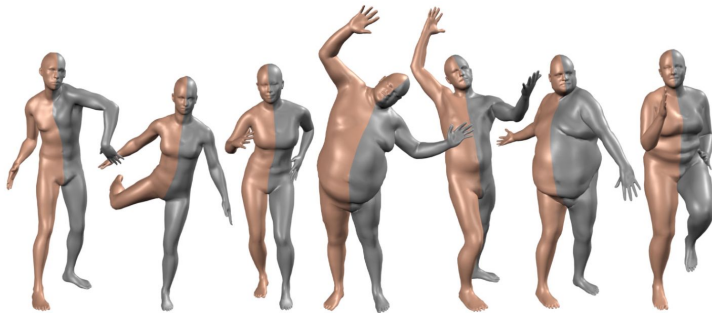
Problem:

- ▶ All humans in similar (but not identical) neutral pose
- ▶ How to model humans at different poses?



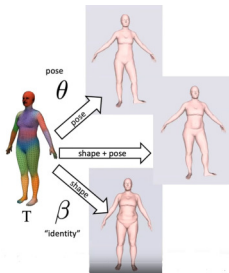
Problem: How to model humans at different poses?

Solution: Skinned Multi-Person Linear (SMPL) model⁴



→ The de-facto standard human model used in research

⁴M. Loper et al. "SMPL: A Skinned Multi-Person Linear Model". In: *ACM Transactions on Graphics (TOG)* 34.6 (2015).



Input:

- Pose parameters
(3 per joint + root orientation)
 $\theta \in \mathbb{R}^{3K+3}$
- Shape parameters
 $\beta \in \mathbb{R}^{10}$

Given: → Pretrained and fixed

- Mean template mesh
(male, female, gender neutral)
with $K = 23$ joints
 $\bar{T} \in \mathbb{R}^{3N}$
- Joint regressor weights
 $\mathcal{J} \in \mathbb{R}^{3K \times 3N}$
- Blend weights
 $\mathcal{W} \in \mathbb{R}^{N \times K}$
- Blend shapes for pose
 $\mathcal{P} \in \mathbb{R}^{3N \times 9K}$
- Blend shapes for shape
 $\mathcal{S} \in \mathbb{R}^{3N \times 10}$

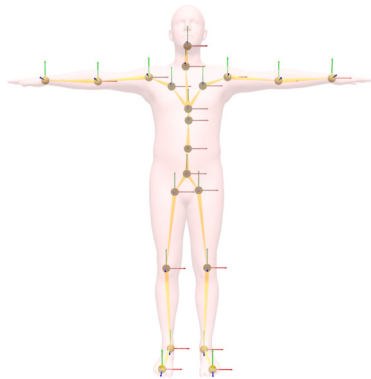


Output:

$$T_{\text{new}}(\beta, \theta) = \text{LBS}(\bar{T}_{S,P}, \mathcal{J}, \theta, \mathcal{W})$$

→ How is this related to the given parameters?

Human model and its pose is defined by a standard skeletal rig (kinematic tree)



$\bar{T}$, θ_{rest}

- ▶ Mean template $\bar{T}$ is an artist-created mesh with $N = 6890$ vertices in rest pose θ_{rest}
- ▶ SMPL model has $K = 23$ joints (+ global orientation):

$$\theta = (\omega_0, \dots, \omega_K)^T \in \mathbb{R}^{3K+3}$$

- ▶ Each part $\omega_k \in \mathbb{R}^3$ is rotation in angle-axis representation relative to its parent



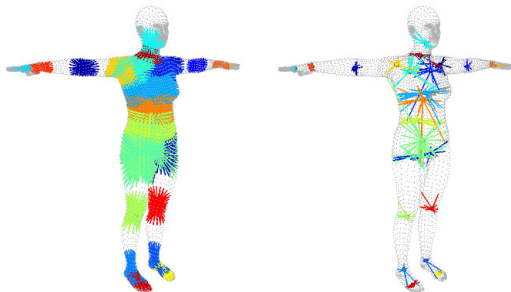
Question: How to get the joint locations?

- ▶ Different body shapes have different joint locations
- ▶ Accurate locations are critical to avoid skinning artifacts when model is posed

Idea: Compute locations from shape-deformed vertices

$$\mathbf{J}(\beta) = \mathcal{J} \cdot (\overline{\mathbf{T}} + \mathbf{B}_S(\beta))$$

- ▶ Matrix $\mathcal{J}$ contains weights to regress/map vertices to locations
- ▶ Only few vertices per joint location sufficient

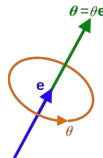


Note that details of $\mathbf{B}_S(\beta)$ will be explained later

Now that we have the pose θ and joint locations $\mathbf{J}$, how can we deform the template mesh?

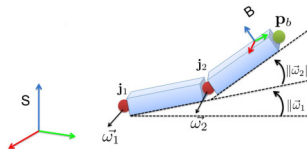
1. Relative transformation of k -th joint:

$$T_k = \left[\begin{array}{c|c} R_{\text{axis-angle}}(\omega_k) & \mathbf{j}_k \\ \hline \mathbf{0} & 1 \end{array} \right]$$



2. Kinematic chain across ancestors $a(k), a^2(k), \dots$ to root:

$$T_k = T_{a(k)} \cdot T_{a^2(k)} \cdot \dots \cdot T_{\text{root}}$$

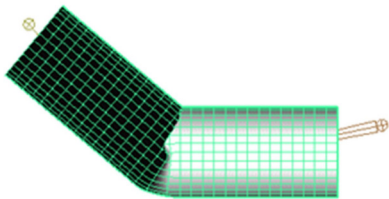


Example: $k = \text{"left knee"}$

$$T_{\text{world} \rightarrow \text{L_Knee}} = T_{\text{L_Hip} \rightarrow \text{L_Knee}} \cdot T_{\text{Pelvis} \rightarrow \text{L_Hip}} \cdot T_{\text{world} \rightarrow \text{Pelvis}}$$

Question: How to apply the transformation to a mesh vertex v_i of the template?

Rigid Skinning



$$\mathbf{v}_i' = \mathbf{T}_{b(i)} \mathbf{v}_i$$

- ▶ Only closest body part $b(i)$ influences vertex
- ▶ Leads to unpleasant discontinuities around joints

Linear Blend Skinning (LBS)



$$\mathbf{v}_i' = \sum_{k=1}^K w_{ik} \mathbf{T}_k \mathbf{v}_i$$

- ▶ All body parts can theoretically influence vertex
- ▶ Weights are usually sparse (e.g. elbow does not influence leg)

Question: How to apply the transformation to a mesh vertex v_i of the template?

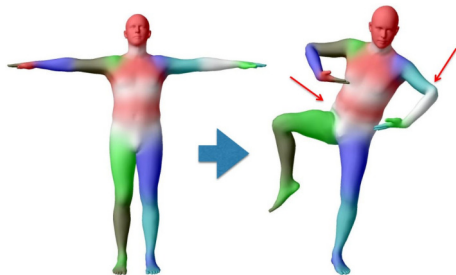
Use Linear Blend Skinning as the **Skinning Function**:

$$\text{LBS}(v_i, \mathbf{J}, \boldsymbol{\theta}, \mathcal{W}) = \sum_{k=1}^K w_{ik} \left(\underbrace{\mathbf{T}_k(\boldsymbol{\theta}, \mathbf{J})}_{\text{world} \rightarrow \text{current}} \cdot \underbrace{\mathbf{T}_k(\boldsymbol{\theta}_{\text{rest}}, \mathbf{J})^{-1}}_{\text{rest} \rightarrow \text{world}} \right) v_i$$

- Transformations are relative to world frame
→ Chain to map vertices v_i in rest pose to current frame
- Blend weight matrix is sparse

$$\mathcal{W} = \begin{pmatrix} w_{ik} \end{pmatrix}$$

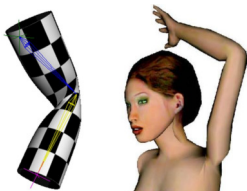
→ 4 joints per vertex with $w_{ik} \neq 0$



→ Suboptimal deformation

Problem: Artifacts by Linear Blend Skinning

"Bow Tie" Effect → Large angles



"Candy Wrapper" Effect → Twist motion



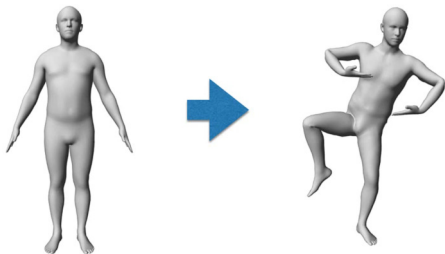
Observation: This "loss in mesh volume" is caused by **blending matrices** of rigid transformations

$$\mathbf{v}_i' = \sum_{k=1}^K w_{ik} \mathbf{T}_k \mathbf{v}_i = \left(\sum_{k=1}^K w_{ik} \mathbf{T}_k \right) \cdot \mathbf{v}_i = \mathbf{T}_{\text{blended}} \cdot \mathbf{v}_i$$

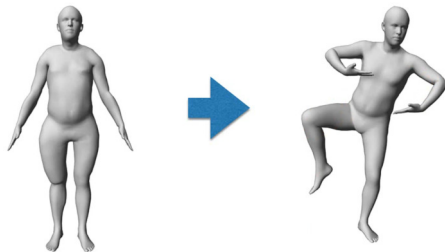
→ Blended matrix is not necessarily rigid body transformation

Problem: Artifacts by Linear Blend Skinning

Template Mesh



Deformed Template Mesh



Idea: Apply pose-dependent deformation to the template before blend skinning

- ▶ "Natural" effects: bulging of muscles, ...
- ▶ Correction of LBS artifacts

Define vertex displacements B_P , also called **blend shapes**, with PCA-like combination of pose θ

$$\begin{aligned} B_P(\theta) &= \sum_{n=1}^{9K} (R_n(\theta) - R_n(\theta_{\text{rest}})) P_n = \left[\underbrace{P_1}_{\in \mathbb{R}^{3N}} \mid \dots \mid P_{9K} \right] \cdot \left(\begin{bmatrix} R_1(\theta) \\ \vdots \\ R_{9K}(\theta) \end{bmatrix} - \begin{bmatrix} R_1(\theta_{\text{rest}}) \\ \vdots \\ R_{9K}(\theta_{\text{rest}}) \end{bmatrix} \right) \\ &= \mathcal{P} \cdot (\mathbf{R}(\theta) - \mathbf{R}(\theta_{\text{rest}})) \end{aligned}$$

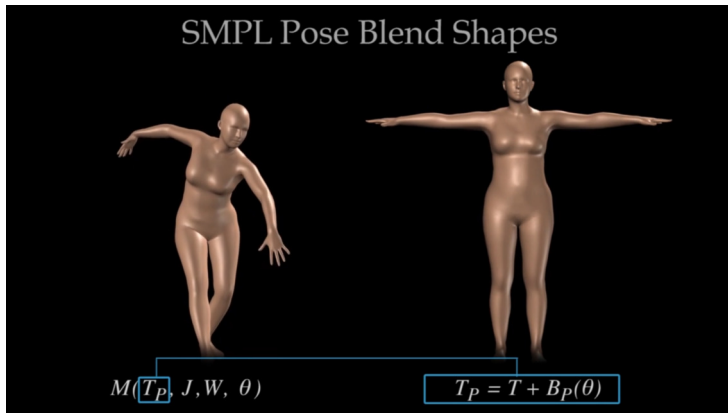
Observations:

- ▶ Subtract rest pose rotations $\mathbf{R}(\theta_{\text{rest}})$ to ensure that contribution is relative to rest pose
- ▶ Use linear combination of sines and cosines from **rotation matrices** instead of **angles**

$$\mathbf{R}(\theta) = \text{vec} \left(\left[\underbrace{(\mathbf{R}_{\text{axis-angle}}(\omega_1), \dots, \mathbf{R}_{\text{axis-angle}}(\omega_K))}_{\in \mathbb{R}^{3 \times 3}} \right]^T \right) \in \mathbb{R}^{9K}$$

→ Same terms that get blended in LBS

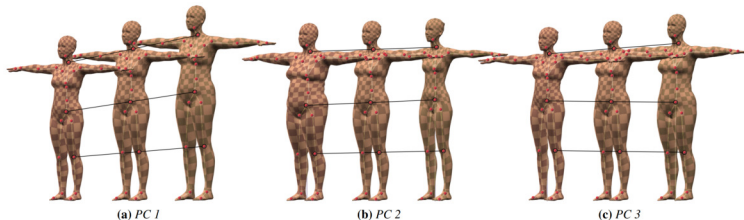
Define vertex displacements B_P , also called **blend shapes**, with PCA-like combination of pose θ



<https://youtu.be/kuB1UyHeV5U?t=161>

Similarly, define vertex displacements/blend shapes B_S with PCA-like combination of shape β

$$B_S(\beta) = \sum_{n=1}^{10} \beta_n S_n = \left[\underbrace{S_1}_{\in \mathbb{R}^{3N}} \mid \dots \mid S_{10} \right] \cdot \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_{10} \end{bmatrix} = S \cdot \beta$$



<https://youtu.be/rzpiSYTrRU0?t=2775>

Summary: Putting all pieces together, the SMPL model is indeed quite "simple"



Mean Template

$$\bar{T}$$



With Shape
Blend Shapes

$$\underbrace{\bar{T}_S(\beta)}_{= \bar{T} + B_S(\beta)}$$

$$J(\beta)$$



With Shape + Pose
Blend Shapes

$$\underbrace{\bar{T}_{S,P}(\beta, \theta)}_{= \bar{T} + B_S(\beta) + B_P(\theta)}$$

$$J(\beta)$$

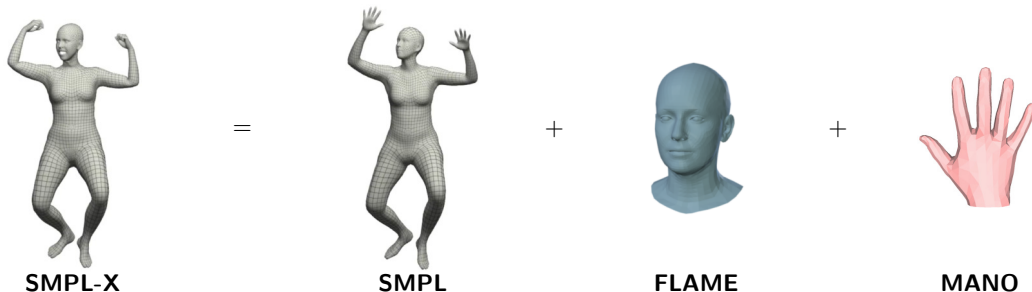


Fully Deformed Model

$$\text{LBS}(\bar{T}_{S,P}, J, \theta, \mathcal{W})$$

Problem: Although SMPL can model humans quite well in shape and pose, they still look "emotionless"

Idea: Use better models for under-represented parts → SMPL-X⁵



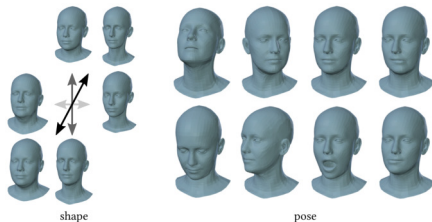
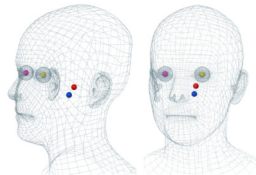
→ **Larger model** $\bar{T}_{S,P,E}(\beta, \theta, \psi)$: 54 joints, $N = 10475$ vertices, shape β , pose θ , expression ψ

⁵G. Pavlakos et al. "Expressive Body Capture: 3D Hands, Face, and Body from a Single Image". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019.

A significant part of the human "identity" is contained in the face $\rightarrow$ FLAME⁶

Same underlying model as SMPL:

- ▶ Artist-generated mean template with $N = 5023$ vertices
- ▶ $K = 4$ joints + global orientation (left eye, right eye, jaw, neck)
- ▶ Blend shapes for Pose $B_P(\theta_{\text{flame}})$
- ▶ Blend shapes for Shape $B_S(\beta_{\text{flame}})$
- ▶ **Additional** blend shapes for **Expression** $B_E(\psi)$ with 100 components



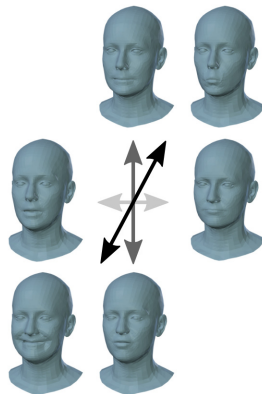
⁶T. Li et al. "Learning a model of facial shape and expression from 4D scans". In: *ACM Transactions on Graphics (TOG)* 36.6 (2017).

Like for pose and shape, use blend shapes $B_E(\psi)$ to model expressions $\psi \in \mathbb{R}^{100}$

$$B_E(\psi) = \sum_{n=1}^{100} \psi_n E_n = \left[\underbrace{E_1}_{\in \mathbb{R}^{3N}} \quad \dots \quad E_{100} \right] \cdot \begin{bmatrix} \psi_1 \\ \vdots \\ \psi_{100} \end{bmatrix} = \mathcal{E} \cdot \psi$$

Example:

<https://youtu.be/36rPTkhiJTM?t=241>

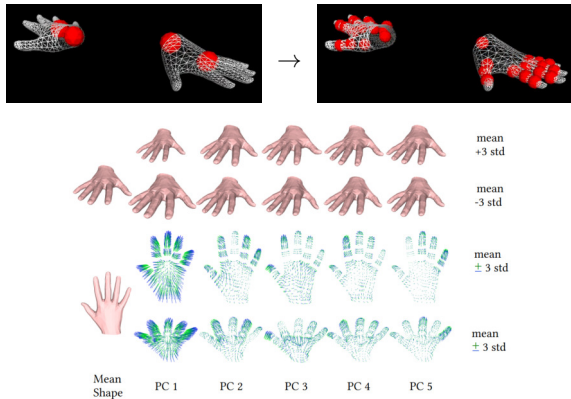


expression

Similar to the face, hands convey a lot of plausibility and realism of humans $\rightarrow$ MANO⁷

Same underlying model as SMPL:

- ▶ Artist-generated mean template with $N = 778$ vertices
- ▶ $K = 15$ joints + global orientation
- ▶ Blend shapes for Shape $B_S(\beta_{\text{mano}})$
 $\rightarrow$ Low-dimensional $\beta_{\text{mano}} \in \mathbb{R}^6$ is sufficient
- ▶ Blend shapes for Pose $B_P(\theta_{\text{mano}})$
 $\rightarrow \theta_{\text{mano}} \in \mathbb{R}^{48}$ is high-dimensional



⁷J. Romero et al. “Embodied Hands: Modeling and Capturing Hands and Bodies Together”. In: *ACM Transactions on Graphics (TOG)* 36.6 (2017).

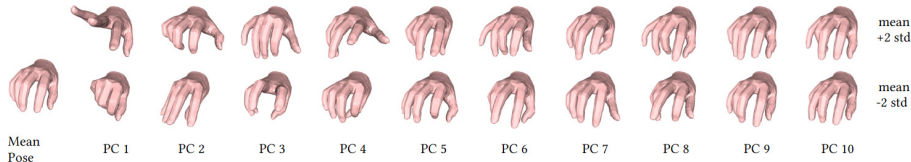
Problem: Adding MANO to SMPL would more than double the amount of joints

$$23 \rightarrow \underbrace{23}_{\text{SMPL}} + 2 \cdot \underbrace{15}_{\text{MANO}} - \underbrace{2}_{\text{replaced joints}} = 51$$

Observation: Most hand joints are restricted to 1 DoF $\rightarrow$ low-dimensional manifold

$$\theta_{\text{mano}} = \sum_{n=1}^6 h_n M_n = \left[\underbrace{M_1}_{\in \mathbb{R}^{48}} \mid \dots \mid M_6 \right] \cdot \begin{bmatrix} h_1 \\ \vdots \\ h_6 \end{bmatrix} = \mathcal{M} \cdot \mathbf{h}$$

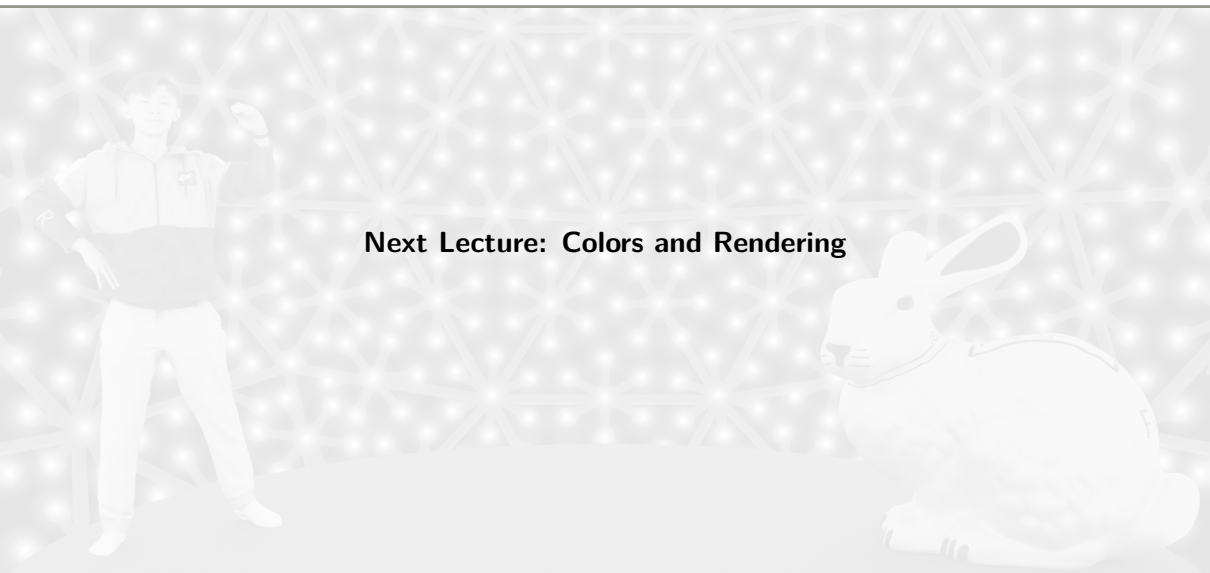
$\rightarrow \mathbf{h} \in \mathbb{R}^6$ sufficient to cover 81 % of variance



Results:



<https://youtu.be/XyXIEmapWkw>



Next Lecture: Colors and Rendering

→ **Colors and Rendering**



- [1] B. Allen, B. Curless, and Z. Popović. "The space of human body shapes: reconstruction and parameterization from range scans". In: *ACM Transactions on Graphics (TOG)* 22.3 (2003).
- [2] V. Blanz and T. Vetter. "A Morphable Model for the Synthesis of 3D Faces". In: *Annual Conference on Computer Graphics and Interactive Techniques*. 1999.
- [3] T. Li, T. Bolkart, M. J. Black, H. Li, and J. Romero. "Learning a model of facial shape and expression from 4D scans". In: *ACM Transactions on Graphics (TOG)* 36.6 (2017).
- [4] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. "SMPL: A Skinned Multi-Person Linear Model". In: *ACM Transactions on Graphics (TOG)* 34.6 (2015).
- [5] R. A. Newcombe, D. Fox, and S. M. Seitz. "DynamicFusion: Reconstruction and Tracking of Non-rigid Scenes in Real-Time". In: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2015.
- [6] G. Pavlakos, V. Choutas, N. Ghorbani, T. Bolkart, A. A. Osman, D. Tzionas, and M. J. Black. "Expressive Body Capture: 3D Hands, Face, and Body from a Single Image". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019.
- [7] J. Romero, D. Tzionas, and M. J. Black. "Embodied Hands: Modeling and Capturing Hands and Bodies Together". In: *ACM Transactions on Graphics (TOG)* 36.6 (2017).